

Question 1: Answer all the questions. (10 Marks)

Rahim and Karim are two friends who are both fascinated by circuits. They have built a circuit as shown in **Figure 1**. During a discussion, Rahim claims that the present condition at terminal **a-b** will ensure maximum power transfer. However, Karim disagrees, stating that the resistance between terminals **a-b** should equal the Thevenin equivalent resistance of the circuit for maximum power transfer. Answer the following questions:

- i) **Determine** the Thevenin equivalent circuit and current through the 8Ω resistor.
- ii) **Explain** who is correct between Rahim and Karim. Also, **determine** the maximum power.

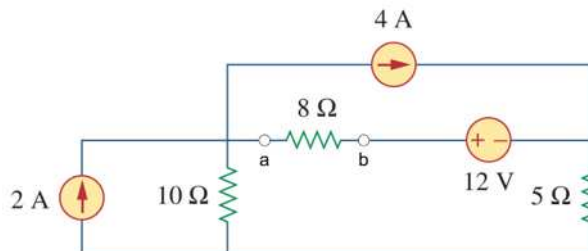


Figure 1

Question 2: Answer all the questions. (10 Marks)

Use the superposition theorem to find the voltage V_x in the circuit shown in **Figure 2**.

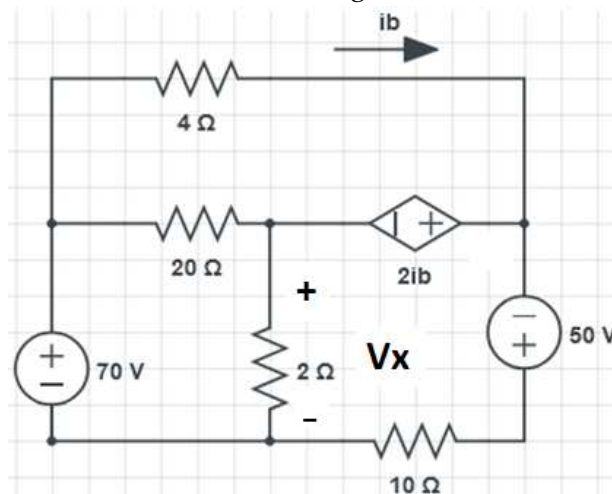


Figure 2

Question 3: Answer all the questions (12 Marks)

In 'Figure 3', the current of the current source is $i_s(t) = 10\cos(4t + 30^\circ)$ A and the other marked current is $i(t) = 4.53\cos(4t - 102.1^\circ)$ A. In the circuit, Z is an unknown impedance. Observe the circuit shown in **Figure 3** and answer the following questions:

- Determine** the voltage across the capacitor 0.125F and then determine current through it.
- Determine** the current through the ' Z ' using KCL at node ' a '.
- Keeping in mind that ' Z ' is parallel to 0.125F capacitor, **determine** the value of ' Z ' using ohm's law.
- Find** the equivalent impedance, Z_{eq} , with respect to the terminals ' a ' and ' b '.

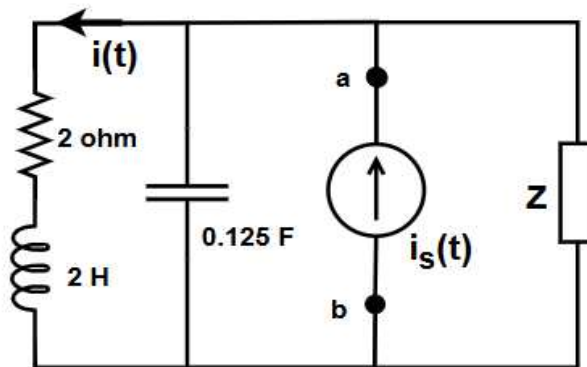


Figure 3

Question 4: Answer all the questions.

(8 Marks)

- Find** the RMS value, I_{RMS} of the current waveform shown in **Figure 4(a)**.
- Use** the RMS value from (i) as the maximum amplitude of the sinusoidal current source in the circuit shown in **Figure 4(b)**. **Determine** i_0 and the **average real power** absorbed by the 3-ohm resistor using the Current Division Rule (CDR), assuming an angular frequency of 100 rad/s .

[4+4]

CO4

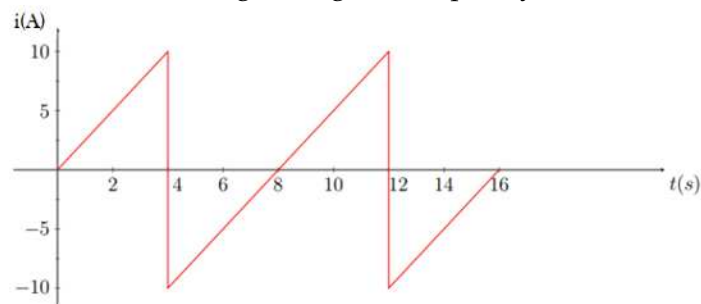


Figure 4(a)

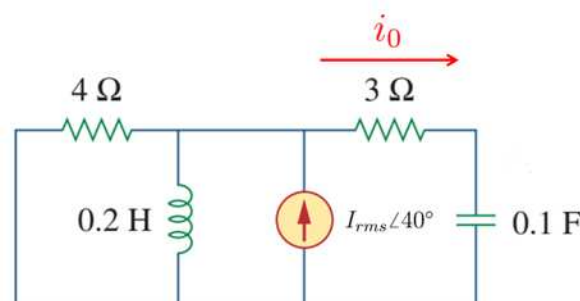


Figure 4(b)